

# Binomial Theorem

## Question1

In the binomial expansion of  $(p - q)^{14}$ , if the sum of 7th term and 8 th term is zero, then  $\frac{p+q}{p-q} =$

AP EAPCET 2025 - 26th May Morning Shift

Options:

A.

14

B.

15

C.

16

D.

13

**Answer: B**

**Solution:**

We are given the expression  $(p - q)^{14}$ . We need to find the value of  $\frac{p+q}{p-q}$ , given that the sum of the 7th and 8th terms in its expansion is zero.

**Finding the 7th and 8th Terms**

In the binomial expansion of  $(p - q)^{14}$ , the formula for the  $(r + 1)$ th term is:  $T_{r+1} = {}^{14}C_r p^{14-r} (-q)^r$ .

The 7th term,  $T_7$ , is  ${}^{14}C_6 p^8 (-q)^6 = {}^{14}C_6 p^8 q^6$ .

The 8th term,  $T_8$ , is  ${}^{14}C_7 p^7 (-q)^7 = -{}^{14}C_7 p^7 q^7$ .

**Setting Up the Equation**

We are told that  $T_7 + T_8 = 0$ .

This means:  ${}^{14}C_6 p^8 q^6 - {}^{14}C_7 p^7 q^7 = 0$ .

We can factor out  $p^7 q^6$  from both terms:  $p^7 q^6 ({}^{14}C_6 p - {}^{14}C_7 q) = 0$ .

Because  $p$  and  $q$  are not zero, we must have  ${}^{14}C_6 p = {}^{14}C_7 q$ .

**Solving for  $\frac{p}{q}$**

Divide both sides by  $q$  and  ${}^{14}C_6$  to get  $\frac{p}{q} = \frac{{}^{14}C_7}{{}^{14}C_6}$ .

Now,  ${}^{14}C_7 = \frac{14!}{7!7!}$  and  ${}^{14}C_6 = \frac{14!}{6!8!}$ .

Write  $\frac{{}^{14}C_7}{{}^{14}C_6}$  as  $\frac{8}{7}$  (after cancelling out common terms, which are shown in the expansion above).

**Express  $p$  and  $q$**

If  $\frac{p}{q} = \frac{8}{7}$ , let  $p = 8k$  and  $q = 7k$  for some  $k$ .

**Find  $\frac{p+q}{p-q}$**

Then  $p + q = 8k + 7k = 15k$  and  $p - q = 8k - 7k = k$ .

So,  $\frac{p+q}{p-q} = \frac{15k}{k} = 15$ .

---

## Question2

**The numerically greatest term in the expansion of  $(x + 3y)^{13}$ , when  $x = \frac{1}{2}$  and  $y = \frac{1}{3}$  is**

### AP EAPCET 2025 - 26th May Morning Shift

**Options:**

A.

$${}^{13}C_9 \left(\frac{1}{3}\right)^4$$

B.

$${}^{13}C_4 \left(\frac{1}{2}\right)^9$$

C.

$${}^{13}C_9 \left(\frac{1}{2}\right)^4$$

D.

$${}^{13}C_{10} \frac{1}{2^4}$$

**Answer: C**

**Solution:**

General term of  $(x + 3y)^{13}$  is

$$T_{r+1} = {}^{13}C_r (x)^{13-r} (3y)^r$$

at  $x = 1/2, y = 1/3$

$$\text{So, } 3y = 1$$

$$\text{then, } T_{r+1} = {}^{13}C_r \left(\frac{1}{2}\right)^{13-r} (1)^r = {}^{13}C_r \left(\frac{1}{2}\right)^{13-r}$$

For maximum term

$$\left| \frac{T_{r+1}}{T_r} \right| = \frac{{}^{13}C_r}{{}^{13}C_{r-1}} \left(\frac{1}{2}\right)^{13-r-(13-(r-1))}$$
$$= \frac{{}^{13}C_r}{{}^{13}C_{r-1}} \left(\frac{1}{2}\right)^{-1} = \frac{14-r}{r} \times 2$$

$$\text{put } \left| \frac{T_{r+1}}{T_r} \right| = 1$$

$$\Rightarrow 28 - 2r = r \Rightarrow r = 28/3 \approx 9.33$$

So, maximum occurs either  $r = 9$  or  $r = 10$

for  $r = 9$

$$T_{10} = {}^{13}C_9 \left(\frac{1}{2}\right)^4 = 44.6875$$

and for  $r = 10$

$$T_{11} = {}^{13}C_{10} \left(\frac{1}{2}\right)^3 = 35.75$$

Hence, the greatest term is  $T_{10}$

$$T_{10} = {}^{13}C_9 \left(\frac{1}{2}\right)^4$$

---

## Question3

**The coefficient of  $x^{10}$  in the expansion of  $\left(x + \frac{2}{x} - 5\right)^{12}$  is**

**AP EAPCET 2025 - 26th May Evening Shift**

**Options:**

A.

B.

2132

C.

1892

D.

862

**Answer: A**

**Solution:**

Coefficient of  $x^{10}$  in the expansion of  $\left(x + \frac{2}{x} - 5\right)^{12}$

Using multinomial theorem

$$= \frac{12!}{r_1!r_2!r_3!} x^{r_1} \left(\frac{2}{x}\right)^{r_2} (-5)^{r_3}$$

for coefficient of  $x^{10}$

$$r_1 + r_2 + r_3 = 12 \Rightarrow r_1 - r_2 = 10$$

| $r_1$ | $r_2$ | $r_3$ |
|-------|-------|-------|
| 10    | 0     | 2     |
| 11    | 1     | 0     |

Coefficient

$$\begin{aligned} &= \frac{12!}{10!0!2!} 2^0 \cdot (-5)^2 + \frac{12!}{11!1!0!} 2^1 \cdot (-5)^0 \\ &= \frac{12 \times 11}{2} \times 25 + 24 \\ &= 1650 + 24 = 1674 \end{aligned}$$

---

## Question4

Let  $S_1 = \sum_{j=1}^{10} j(j-1) \cdot {}^{10}C_j$ ,  $S_2 = \sum_{j=1}^{10} j \cdot {}^{10}C_j$  and

$$S_3 = \sum_{j=1}^{10} j^2 \cdot {}^{10}C_j$$

**Assertion (A)**  $S_3 = 55 \times 2^9$

**Reason (R)**  $S_1 = 90 \times 2^8$  and  $S_2 = 10 \times 2^8$

## AP EAPCET 2025 - 26th May Evening Shift

### Options:

A.

Both (A) and (R) are true and R is the correct explanation of (A)

B.

Both (A) and (R) are true but (R) is not the correct explanation of (A)

C.

(A) is true, but (R) is false

D.

(A) is false, but (R) is true

**Answer: C**

### Solution:

Given,

$$\begin{aligned} S_1 &= \sum_{j=1}^{10} j(j-1) \cdot {}^{10}C_j \\ &= \sum_{j=1}^{10} j(j-1) \frac{10}{j} \cdot \frac{9}{j-1} \cdot {}^8C_{j-2} \\ &= \sum_{j=2}^{10} 90 \cdot {}^8C_{j-2} = 90 \cdot 2^8 \end{aligned}$$

$$\begin{aligned} \text{and } S_2 &= \sum_{j=1}^{10} j \cdot {}^{10}C_j \\ &= \sum_{j=1}^{10} j \cdot \frac{10}{j} {}^9C_{j-1} = 10 \sum_{j=1}^{10} {}^9C_{j-1} \\ &= 10 \cdot 2^9 \\ S_3 &= \sum_{j=1}^{10} j^2 {}^{10}C_j = 10 \sum_{j=1}^{10} j \cdot {}^9C_{j-1} \\ &= 10 \cdot 11 \cdot 2^{9-1} = 110 \cdot 2^8 \\ &\quad \left[ \because \sum r \cdot {}^nC_r = n \cdot 2^{n-1} \right] \\ &= 55 \cdot 2^9 \end{aligned}$$

---

## Question5

If  $y = \frac{3}{4} + \frac{3 \cdot 5}{4 \cdot 8} + \frac{3 \cdot 5 \cdot 7}{4 \cdot 8 \cdot 12} + \dots + \infty$ , then

**AP EAPCET 2025 - 24th May Morning Shift**

**Options:**

A.

$$y^2 - 2y + 5 = 0$$

B.

$$y^2 + 2y - 7 = 0$$

C.

$$y^2 - 3y + 4 = 0$$

D.

$$y^2 + 4y - 6 = 0$$

**Answer: B**

**Solution:**

Given,

$$\begin{aligned} y &= \frac{3}{4} + \frac{3 \cdot 5}{4 \cdot 8} + \frac{3 \cdot 5 \cdot 7}{4 \cdot 8 \cdot 12} + \dots + \infty \\ &= \frac{3}{1!} \cdot \frac{1}{4} + \frac{3 \cdot 5}{2!} \cdot \frac{1}{4^2} + \frac{3 \cdot 5 \cdot 7}{3!} \cdot \frac{1}{4^3} + \dots + \infty \end{aligned}$$

Since, using binomial expansion

$$(1 - x)^{-n} = 1 + nx + \frac{n(n+1)}{2!}x^2 + \frac{n(n+1)(n+2)}{3!}x^3 + \dots$$

$$\text{Here, } nx = \frac{3}{4} \quad \dots (i)$$

$$\frac{n(n+1)}{2}x^2 = \frac{3 \cdot 5}{2!} \cdot \frac{1}{4^2}$$

$$\Rightarrow n(n+1)x^2 = \frac{15}{16} \quad \dots (ii)$$

From Eq. (i) and (ii), we get

$$\begin{aligned} \frac{n(n+1)x^2}{n^2x^2} &= \frac{15/16}{(3/4)^2} \\ &= \frac{15/16}{9/16} = \frac{5}{3} \end{aligned}$$

$$\Rightarrow \frac{n+1}{n} = \frac{5}{3}$$

$$\Rightarrow 3n + 3 = 5n$$

$$\Rightarrow n = \frac{3}{2}$$

$$\text{So, } x = \frac{3}{4} \cdot \frac{1}{n} = \frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$$

$$\begin{aligned} \therefore y &= (1-x)^{-n} - 1 = \left(1 - \frac{1}{2}\right)^{-3/2} - 1 \\ &= \left(\frac{1}{2}\right)^{-3/2} - 1 = 2^{3/2} - 1 = 2\sqrt{2} - 1 \end{aligned}$$

$$\text{So, } y + 1 = 2\sqrt{2} - 1 + 1 = 2\sqrt{2}$$

$$(y+1)^2 = (2\sqrt{2})^2 = 8$$

$$\Rightarrow y^2 + 2y + 1 - 8 = 0$$

$$\Rightarrow y^2 + 2y - 7 = 0$$

Thus, the quadratic equation is

$$y^2 + 2y - 7 = 0$$

## Question6

**Sum of the coefficients of  $x^4$  and  $x^6$  in the expansion of  $(1 + x - x^2)^6$  is**

**AP EAPCET 2025 - 24th May Morning Shift**

**Options:**

A.

121

B.

-91

C.

11

D.

31

**Answer: C**

**Solution:**

We will expand  $(1 + x - x^2)^6$  and find the coefficients of  $x^4$  and  $x^6$ .

Multinomial expansion states that

$$(a_1 + a_2 + a_3)^n = \sum_{a_1^{k_1} a_2^{k_2} a_3^{k_3}} \frac{n!}{k_1! k_2! k_3!}$$

where  $k_1 + k_2 + k_3 = n$

In our case,  $a_1 = 1, a_2 = x, a_3 = -x^2$  and  $n = 6$

From  $k_1 + k_2 + k_3 = 6$

$$\Rightarrow k_1 = 6 - k_2 - k_3 \quad \dots (i)$$

To get  $x^4$ , we can have different combinations of  $x$  and  $x^2$

$k_1$  : number of times 1 is chosen

$k_2$  : number of times  $x$  is chosen

$k_3$  : number of times  $x^2$  is chosen

The equation is:  $k_2 + 2k_3 = 4 \quad \dots (ii)$

Now, if  $k_3 = 0$ , then  $k_2 = 4$  and  $k_1 = 2$

If  $k_3 = 1$ , then  $k_2 = 2$  and  $k_1 = 3$

If  $k_3 = 2$ , then,  $k_2 = 0, k_1 = 4$

Now, find the coefficients for each valid combination.

Coefficient of

$$\begin{aligned} x^4 &= \frac{6!}{2!4!0!} - \frac{6!}{3!2!1!} + \frac{6!}{4!0!2!} \\ &= 15 - 60 + 15 = -30 \end{aligned}$$

Now, for  $x^6, k_2 + 2k_3 = 6$  and

$$k_1 + k_2 + k_3 = 6$$

If  $k_3 = 0$ , then  $k_2 = 6$ , and  $k_1 = 0$

If  $k_3 = 1$ , then  $k_2 = 4$ , and  $k_1 = 1$

If  $k_3 = 2$  then  $k_2 = 2$  and  $k_1 = 2$

If  $k_3 = 3$ , then  $k_2 = 0$ , and  $k_1 = 3$

$$\begin{aligned} \text{So, coefficient of } x^6 &= \frac{6!}{0!6!0!} - \frac{6!}{1!4!1!} + \frac{6!}{2!2!2!} - \frac{6!}{3!0!3!} \\ &= 1 - 30 + 90 - 20 = 41 \end{aligned}$$

$$\therefore \text{Sum of the coefficient of } x^4 \text{ and } x^6 = 41 - 30 = 11$$

## Question7

If  $11^{12} - 11^2 = k(5 \times 10^9 + 6 \times 10^9 + 33 \times 10^8 + 110 \times 10^7 + \dots + 33)$ , then  $k =$

## AP EAPCET 2025 - 23rd May Evening Shift

Options:

A.

20

B.

50

C.

100

D.

200

**Answer: D**

**Solution:**

Given,

$$11^{12} - 11^2 = k(5 \times 10^9 + 6 \times 10^9 + 33 \times 10^8 + 110 \times 10^7 + \dots + 33)$$

$$\begin{aligned} 11^{12} - 11^2 &= (10 + 1)^{12} - (10 + 1)^2 \\ &= 10^{12} + 12 \times 10^{11} + 66 \times 10^{10} + 220 \times 10^9 \dots + 66 \times 10^2 + 120 + 1 - 121 \end{aligned}$$

$$= 10^{12} + 12 \times 10^{11} + 66 \times 10^{10} + 220 \times 10^9 \dots + 66 \times 10^2$$

$$= 2 \times 10^2 (5 \times 10^9 + 6 \times 10^9 + 33 \times 10^8 + 110 \times 10^7 + \dots + 33)$$

$$= 200 (5 \times 10^9 + 6 \times 10^9 + 33 \times 10^8 + 110 \times 10^7 + \dots + 33)$$

$$\therefore k = 200$$

---

## Question8

If  $C_0, C_2, \dots, C_n$  are the binomial coefficients in the expansion of  $(1 + x)^n$ , then

$$(C_0 + C_1) - (C_2 + C_3) + (C_4 + C_5) - (C_6 + C_7) + \dots =$$

## AP EAPCET 2025 - 23rd May Evening Shift

Options:

A.

$$2^{n/2} \left( \cos \frac{n\pi}{4} + i \sin \frac{n\pi}{4} \right)$$

B.

$$2^{n/2} \left( \cos \frac{n\pi}{3} + i \sin \frac{n\pi}{3} \right)$$

C.

$$2^{n/2} \left( \cos \frac{n\pi}{3} + i \sin \frac{n\pi}{3} \right)$$

D.

$$2^{n/2} \left( \cos \frac{n\pi}{4} + \sin \frac{n\pi}{4} \right)$$

**Answer: D**

**Solution:**

$$\because (1+x)^n = \sum_{k=0}^n C_k x^k$$

and

$$(C_0 + C_1) - (C_2 + C_3) + (C_4 + C_5) - \dots$$

can be grouped as

$$\sum_{k=0}^{n/2} (-1)^k (C_{2k} + C_{2k+1})$$

$$\text{and } (1+i)^n = \sum_{k=0}^n C_k i^k$$

$$\operatorname{Re} [(1+i)^n] = C_0 - C_2 + C_4 - C_6 + \dots$$

$$\operatorname{Im} [(1+i)^n] = C_1 - C_3 + C_5 - C_7 + \dots$$

Adding both

$$\begin{aligned} \operatorname{Re} [(1+i)^n] + \operatorname{Im} [(1+i)^n] \\ = (C_0 + C_1) - (C_2 + C_3) + (C_4 + C_5) - \dots \end{aligned}$$

Now,  $(1+i)^n$

$$= (\sqrt{2})^n \left( \cos \left( \frac{n\pi}{4} \right) + i \sin \left( \frac{n\pi}{4} \right) \right)$$

$$\text{Hence, } (C_0 + C_1) - (C_2 + C_3) + (C_4 + C_5) - \dots$$

$$\begin{aligned} &= \operatorname{Re} [(1+i)^n] + \operatorname{Im} [(1+i)^n] \\ &= 2^{n/2} \left( \cos \left( \frac{n\pi}{4} \right) + \sin \left( \frac{n\pi}{4} \right) \right) \end{aligned}$$

-----

## Question9

The mean and variance of a binomial distribution are  $x$  and 5 respectively. If  $x$  is an integer, then the possible values for  $x$  are

AP EAPCET 2025 - 23rd May Evening Shift

Options:

A.

6, 10, 30

B.

8, 12, 28

C.

10, 15, 25

D.

9, 18, 24

**Answer: A**

**Solution:**

The mean of a binomial distribution is  $x$ , and the variance is 5.

We know that the mean,  $\mu = np = x$ . This means  $n$  times  $p$  is  $x$ .

The formula for variance is  $\sigma^2 = np(1 - p) = 5$ . This means  $n$  times  $p$  times  $(1 - p)$  is 5.

From the mean formula,  $p = \frac{x}{n}$ .

Substitute  $p = \frac{x}{n}$  into the variance formula:

$$x \left(1 - \frac{x}{n}\right) = 5$$

If we open the bracket, we get:  $x - \frac{x^2}{n} = 5$

Take  $\frac{x^2}{n}$  to the other side:  $\frac{x^2}{n} = x - 5$

Now, solve for  $n$ :  $n = \frac{x^2}{x-5}$

To get a whole number for  $n$ ,  $x - 5$  must divide  $x^2$  exactly. Let's check different  $x$  values:

If  $x = 6$ :  $n = \frac{36}{1} = 36$  is an integer.

If  $x = 10$ :  $n = \frac{100}{5} = 20$  is an integer.

If  $x = 30$ :  $n = \frac{900}{25} = 36$  is an integer.

So, the possible values for  $x$  are: 6, 10, 30.

---

## Question10

If the coefficients of  $x^{10}$  and  $x^{11}$  in the expansion of  $(1 + \alpha x + \beta x^2)(1 + x)^{11}$  are 396 and 144 respectively, then  $\alpha^2 + \beta^2 =$

**AP EAPCET 2025 - 23rd May Morning Shift**

**Options:**

A.

10

B.

13

C.

25

D.

20

**Answer: B**

**Solution:**

As we know that, coefficient of  $x^{11}$  in the expansion of  $(1 + x)^{11}$  is  ${}^{11}C_{11}$ .

Now, we have to find coefficient of  $x^{11}$  in the expansion of  $(1 + \alpha x + \beta x^2)(1 + x)^{11}$ . When we choosing 1 from  $(1 + \alpha x + \beta x^2)$  then we need  $x^{11}$  from  $(1 + x)^{11}$ .

So, coefficient =  ${}^{11}C_{11} = 1$

Similarly, we choosing  $\alpha x$  from  $(1 + \alpha x + \beta x^2)$  then we need  $x^{10}$  from  $(1 + x)^{11}$  = Coefficient =  $\alpha \times {}^{11}C_{10} = 11\alpha$   
and when we choosing  $\beta x^2$  from  $(1 + \alpha x + \beta x^2)$  then we need  $x^9$  from  $(1 + x)$  = Coefficient =  $\beta \times {}^{11}C_9 = 55 \cdot \beta$

$\therefore$  Coefficient of  $x^{11}$  in the expansion of  $(1 + \alpha x + \beta x^2)$

$$(1 + x)^{11} = 1 + 11\alpha + 55\beta = 144 \text{ ( given )} \quad \dots(i)$$

Just, in similar way, we have to find coefficient of  $x^{10}$  such that

$$11 + 55\alpha + 165\beta = 396 \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get;

$$\alpha = -2 \text{ and } \beta = 3$$

$$\therefore \alpha^2 + \beta^2 = (-2)^2 + (3)^2 = 13$$


---

## Question11

If  $-\frac{2}{3} < x < \frac{2}{3}$ , then the value of the 5 th term in the expansion of  $\frac{1}{\sqrt[3]{2-3x}}$  when  $x = \frac{1}{2}$  is

### AP EAPCET 2025 - 23rd May Morning Shift

Options:

A.

$$\frac{35}{256(\sqrt[3]{2})}$$

B.

$$\frac{35}{768(\sqrt[3]{2})}$$

C.

$$\frac{7}{768(\sqrt[3]{2})}$$

D.

$$\frac{105}{256(\sqrt[3]{2})}$$

**Answer: B**

**Solution:**

$$\text{Given, } -\frac{2}{3} < x < \frac{2}{3}$$

$$\Rightarrow -2 < 3x < 2 \Rightarrow -1 < \frac{3x}{2} < 1$$

$$\text{Now, } \frac{1}{\sqrt[3]{2-3x}} = \frac{1}{(2-3x)^{1/3}}$$

$$= \frac{1}{2^{1/3}} \left(1 - \frac{3x}{2}\right)^{-1/3}$$

$$= \frac{1}{\sqrt[3]{2}} \left[ 1 + \left(\frac{1}{3}\right) \left(\frac{3x}{2}\right) + \frac{\left(\frac{1}{3}\right)\left(\frac{1}{3}+1\right)}{2 \times 1} \left(\frac{3x}{2}\right)^2 + \frac{\left(\frac{1}{3}\right)\left(\frac{1}{3}+1\right)\left(\frac{1}{3}+2\right)}{3 \times 2 \times 1} \left(\frac{3x}{2}\right)^3 + \frac{\frac{1}{3}\left(\frac{1}{3}+1\right)\left(\frac{1}{3}+2\right)\left(\frac{1}{3}+3\right)}{4 \times 3 \times 2 \times 1} \left(\frac{3x}{2}\right)^4 + \dots \right]$$

$$\therefore 5 \text{ th term when } x = \frac{1}{2} \text{ is given by}$$

$$\frac{1}{\sqrt[3]{2}} \left[ \frac{1}{3} \times \frac{4}{3} \times \frac{7}{3} \times \frac{10}{3} \times \frac{1}{24} \times \frac{81}{16} \times \frac{1}{16} \right]$$

$$= \frac{1}{3\sqrt{2}} \left[ \frac{7 \times 5}{24 \times 2 \times 6} \right] = \frac{35}{768(\sqrt[3]{2})}$$


---

## Question12

The terms containing  $x^r y^s$  (for certain  $r$  and  $s$ ) are present in both the expansions of  $(x + y^2)^{13}$  and  $(x^2 + y)^{14}$ . If  $\alpha$  is the number of such terms, then the sum  $\alpha \sum_{r,s} (r + s) =$

**AP EAPCET 2025 - 22nd May Evening Shift**

**Options:**

A.

27

B.

40

C.

18

D.

35

**Answer: C**

**Solution:**

General term for  $(x + y^2)^{13}$

$$= {}^{13}C_n x^{13-n} y^{2n}$$

The general term for  $(x^2 + y)^{14}$

$$= {}^{14}C_{r_2} x^{2r_2} y^{14-r_2}$$

Equate the power of  $x$  and  $y$

$$13 - r_1 = 2r_2$$

$$2r_1 = 14 - r_2$$

Solving these equation, we get

$$r_1 = 5, r_2 = 4$$

The common term is  $x^{13-5}y^{2(9)} = x^8y^{10}$  The common term is  $x^8y^{10}$

$$\Rightarrow r = 8, s = 10$$

$\alpha = 1$ , since there is only one common term

$$\Rightarrow \alpha(r + s) = 1(8 + 10) = 18$$

$$\therefore \text{Sum } \alpha \sum_{r,s} (r + s) = 18$$

---

## Question13

The coefficient of  $x^3$  in the power series expansion of  $\frac{1+4x-3x^2}{(1+3x)^3}$  is

### AP EAPCET 2025 - 22nd May Evening Shift

Options:

A.

-27

B.

27

C.

153

D.

-153

**Answer: A**

**Solution:**

$$\frac{1+4x-3x^2}{(1+3x)^3} = (1 + 4x - 3x^2)(1 + 3x)^{-3}$$

$$\begin{aligned}
\Rightarrow (1+3x)^{-3} &= 1 - 3(3x) + \frac{(-3)(-4)(3x)^2}{2!} \\
&\quad + \frac{(-3)(-4)(-5)(3x)^3}{3!} + \dots \\
\Rightarrow (1+3x)^{-3} &= 1 - 9x + 54x^2 - 270x^3 + \dots \\
(1+4x-3x^2)(1-9x+54x^2-270x^3+\dots) \\
&= 1(-270x^3) + 4x(54x^2) - 3x^2(-9x) \\
&= -270x^3 + 216x^3 + 27x^3 \\
&= -270 + 216 + 27 = -27
\end{aligned}$$

$\Rightarrow$  The coefficient of  $x^3 = -27$

---

## Question14

If  $k$  is a positive integer and  $10^k$  is a divisor of the number  $9^{11} + 11^9$ , then the greatest value of  $k$  is

### AP EAPCET 2025 - 22nd May Morning Shift

Options:

A.

1

B.

2

C.

3

D.

4

**Answer: B**

**Solution:**

$$\begin{aligned}
9^{11} + 11^9 &= (10-1)^{11} + (10+1)^9 \\
&= {}^{11}C_0 10^{11} + {}^{11}C_1 10^9 + \dots + {}^{11}C_{11} \\
&\quad + {}^9C_0 10^{10} + {}^9C_1 10^8 + \dots + {}^9C_9
\end{aligned}$$

Sum of last two terms of both expansion as other term have at least  $10^2$  as a factor

$$(110 - 1) + 90 + 1 = 200$$

$$\therefore 9^{11} + 11^9 = \text{Multiple of}$$

$$100 + 200 = \text{Multiple of } 100$$

$$\text{Since } 200 = 2 \times 10^2$$

$$\text{So, } 9^{11} + 11^9 \text{ is divisible by } 10^2$$

$$\therefore k = 2$$


---

## Question15

**The number of all possible values of  $k$  for which the expansion  $(\sqrt{x} + \sqrt[k]{y})^{10}$  will have exactly nine irrational terms is**

**AP EAPCET 2025 - 22nd May Morning Shift**

**Options:**

A.

3

B.

4

C.

5

D.

6

**Answer: C**

**Solution:**

$$\text{Given expansion } \left(x^{\frac{1}{2}} + y^{\frac{1}{k}}\right)^{10}$$

$$\text{General term} = {}^{10}C_r x^{\frac{10-r}{2}} y^{\frac{r}{k}}$$

Given expansion have only 9 irrational term

$\therefore$  Expansion has only two rational term Two rational term is possible

LCM of 2 and  $k$  is 10

$$\therefore k = 5$$


---

## Question16

Coefficient of  $x^2$  in the expansion of  $(x^2 + x - 2)^5$  is

AP EAPCET 2025 - 21st May Evening Shift

Options:

A.

800

B.

756

C.

0

D.

512

**Answer: C**

**Solution:**

$$(x^2 + x - 2)^5$$

Apply multinomial theorem

$$\Sigma \frac{5!}{p!q!r!} (x^2)^p (x)^q (-2)^r$$

Such that  $p + q + r = 5$  and  $p, q, r$  are non-negative integer.

$$2p + q = 2$$

$$(p, q, r) = (1, 0, 4)(0, 2, 3)$$

$$\frac{5!}{4!} \left( -4^4 + \frac{5!}{2!3!} (-2^3 \right.$$

$$5 \times 16 + 10(-8) = 80 - 80 = 0$$

---

## Question17

If  $P_n$  denotes the product of the binomial coefficients in the expansion of  $(1 + x)^n$ , then  $\frac{P_{n+1}}{P_n} =$

## AP EAPCET 2025 - 21st May Evening Shift

Options:

A.

$$\frac{n+1}{n!}$$

B.

$$\frac{n^n}{n!}$$

C.

$$\frac{(n+1)^n}{(n+1)!}$$

D.

$$\frac{(n+1)^{n+1}}{(n+1)!}$$

**Answer: D**

**Solution:**

We have,  $P_n = {}^nC_0 \cdot {}^nC_1 {}^nC_2 \dots {}^nC_n$  and

$$\begin{aligned} P_{n+1} &= {}^{n+1}C_0 \cdot {}^{n+1}C_1 {}^{n+1}C_2 \dots {}^{n+1}C_{n+1} \\ \therefore \frac{P_{n+1}}{P_n} &= \frac{{}^{n+1}C_0 \cdot {}^{n+1}C_1 {}^{n+1}C_2 \dots {}^{n+1}C_{n+1}}{{}^nC_0 \cdot {}^nC_1 \cdot {}^nC_2 \dots {}^nC_n} \\ &= \left( \frac{{}^{n+1}C_1}{{}^nC_0} \right) \left( \frac{{}^{n+1}C_2}{{}^nC_1} \right) \left( \frac{{}^{n+1}C_3}{{}^nC_2} \right) \dots \left( \frac{{}^{n+1}C_{n+1}}{{}^nC_n} \right) \\ &= \left( \frac{n+1}{1} \right) \left( \frac{n+1}{2} \right) \left( \frac{n+1}{3} \right) \dots \left( \frac{n+1}{n+1} \right) \\ &= \frac{(n+1)^{n+1}}{(n+1)!} \end{aligned}$$

---

## Question18

The coefficient of  $x^3$  in the expansion of  $\frac{x^4+1}{(x^2+1)(x-1)}$  when it is expressed in terms of positive integral powers of  $x$ , is

**Options:**

A.

0

B.

1

C.

16

D.

24

**Answer: A**

**Solution:**

$$\begin{aligned}\text{Given, } & \frac{x^4+1}{(x^2+1)(x-1)} \\ &= \frac{x^4 \left(1 + \frac{1}{x^4}\right)}{x^3 \left(1 + \frac{1}{x^2}\right) \left(1 - \frac{1}{x}\right)} \\ &= x \left(1 + \frac{1}{x^4}\right) \left(1 + \frac{1}{x^2}\right)^{-1} \left(1 - \frac{1}{x}\right)^{-1} \\ &= x \left(1 + \frac{1}{x^4}\right) \left(1 - \frac{1}{x^2}\right) \left(1 + \frac{1}{x} + \frac{1}{x^2} + \dots\right)\end{aligned}$$

Clearly, there is no terms containing  $x^3$  coefficient of  $x^3 = 0$

-----

## Question19

If  $(1+x)^n = \sum_{r=0}^n C_r x^r$ , then the value of  $C_0 + (C_0 + C_1) + (C_0 + C_1 + C_2) + \dots + (C_0 + C_1 + C_2 + \dots + C_n)$  is

**AP EAPCET 2025 - 21st May Morning Shift**

**Options:**

A.

$nR^{n-1}$

B.

$$2^n + n$$

C.

$$(n+2)2^n$$

D.

$$(n+2)2^{n-1}$$

**Answer: D**

**Solution:**

The given sum can be rewritten as

$$\sum_{k=0}^n \sum_{r=0}^k C_r$$

Now, changing order of summation

$$\sum_{r=0}^n \sum_{k=r}^n C_r \quad \left[ \because \sum_{r=0}^n C_r = 2^n \right]$$

$$\sum_{r=0}^n C_r \sum_{k=r}^n 1 = \sum_{r=0}^n C_r (n - r + 1)$$

$$\begin{aligned} &\Rightarrow (n+1) \sum_{r=0}^n C_r - \sum_{r=0}^n r \cdot C_r \\ &= (n+1)2^n - \sum_{r=0}^n r \cdot {}^{n-1}C_{r-1} \times \frac{n}{r} \\ &= (n+1)2^n - n \sum_{r=0}^n {}^{n-1}C_{r-1} \\ &= (n+1)2^n - n \cdot 2^{n-1} \\ &= n \cdot 2^n + 2^n - n \cdot 2^{n-1} \\ &= 2^{n-1}(n+2) \end{aligned}$$

## Question 20

If  $x$  is so large that terms containing  $x^{-3}, x^{-4}, x^{-5}, \dots$  can be neglected, then the approximate value of  $\left( \frac{3x-5}{4x^2+3} \right)^{-1/5}$  is

**AP EAPCET 2025 - 21st May Morning Shift**

**Options:**

A.

$$\left(\frac{3}{4x}\right)^{4/5} \left(1 - \frac{4}{3x} - \frac{7}{5x^2}\right)$$

B.

$$\left(\frac{4x}{3}\right)^{4/5} \left(1 + \frac{4}{3x} + \frac{13}{5x^2}\right)$$

C.

$$\left(\frac{4x}{3}\right)^{4/5} \left(1 + \frac{4}{3x} - \frac{13}{5x^2}\right)$$

D.

$$\left(\frac{3}{4x}\right)^{4/5} \left(1 - \frac{4}{3x} + \frac{7}{5x^2}\right)$$

**Answer: B**

**Solution:**

$$\begin{aligned} \text{We have, } \left(\frac{3x-5}{4x^2+3}\right)^{-\frac{4}{5}} &= \left(\frac{4x^2+3}{3x-5}\right)^{\frac{4}{5}} \\ &= \left\{ \frac{4x^2 \left(1 + \frac{3}{4x^2}\right)}{3x \left(1 - \frac{5}{3x}\right)} \right\}^{\frac{4}{5}} = \left(\frac{4x}{3}\right)^{\frac{4}{5}} \frac{\left(1 + \frac{3}{4x^2}\right)^{4/5}}{\left(1 - \frac{5}{3x}\right)^{4/5}} \\ &\Rightarrow \left(\frac{4x}{3}\right)^{\frac{4}{5}} \left(1 + \frac{3}{4x^2}\right)^{\frac{4}{5}} \cdot \left(1 - \frac{5}{3x}\right)^{-\frac{4}{5}} \\ &= \left(\frac{4x}{3}\right)^{\frac{4}{5}} \left(1 + \frac{4}{5} \cdot \frac{3}{4x^2} + \dots\right) \\ &\quad \left(1 + \frac{5}{3x} \cdot \frac{4}{5} + \frac{4}{5} \left(+\frac{9}{5}\right) \left(\frac{5}{3x}\right)^2 \frac{1}{2!} + \dots\right) \\ &= \left(\frac{4x}{3}\right)^{\frac{4}{5}} \left(1 + \frac{3}{5x^2} + \dots\right) \left(1 + \frac{4}{3x} + \frac{2}{x^2}\right) \\ &= \left(\frac{4x}{3}\right)^{\frac{4}{5}} \left(1 + \frac{4}{3x} + \left(\frac{3}{5} + 2\right) \frac{1}{x^2}\right) \\ &= \left(\frac{4x}{3}\right)^{\frac{4}{5}} \left(1 + \frac{4}{3x} + \frac{13}{5x^2}\right) \end{aligned}$$

## Question21

The independent term in the expansion of  $(1+x+2x^2)\left(\frac{3x^2}{2}-\frac{1}{3x}\right)^9$  is

**AP EAPCET 2024 - 23th May Morning Shift**

**Options:**

A.  $\frac{18}{7}$

B.  $\frac{7}{18}$

C.  $-\frac{7}{18}$

D.  $-\frac{18}{7}$

**Answer: B**

**Solution:**

$$\begin{aligned}
 &\text{We have, } (1 + x + 2x^2) \left( \frac{3x^2}{2} - \frac{1}{3x} \right)^9 \\
 &= (1 + x + 2x^2) \\
 &\left\{ \sum_{r=0}^9 {}^9C_r \left( \frac{3}{2}x^2 \right)^{9-r} \left( -\frac{1}{3x} \right)^r \right\} \\
 &= (1 + x + 2x^2) \\
 &\left\{ \sum_{r=0}^9 {}^9C_r \left( \frac{3}{2} \right)^{9-r} x^{18-2r} \left( -\frac{1}{3} \right)^r x^{-r} \right\} \\
 &= (1 + x + 2x^2) \\
 &\left\{ \sum_{r=0}^9 {}^9C_r \left( \frac{3}{2} \right)^{9-r} \left( -\frac{1}{3} \right)^r x^{18-3r} \right\} \\
 &= \sum_{r=0}^9 {}^9C_r \left( \frac{3}{2} \right)^{9-r} \left( -\frac{1}{3} \right)^r x^{18-3r} \\
 &\quad + \sum_{r=0}^9 {}^9C_r \left( \frac{3}{2} \right)^{9-r} \left( -\frac{1}{3} \right)^r x^{19-3r} \\
 &\quad + 2 \left\{ \sum_{r=0}^9 {}^9C_r \left( \frac{3}{2} \right)^{9-r} \left( -\frac{1}{3} \right)^r x^{20-3r} \right\}
 \end{aligned}$$

Clearly, first term independent of  $x$  by

$$18 - 3r = 0 \Rightarrow r = 6$$

$\therefore$  Coefficient of independent of  $x$  is

$$\begin{aligned}
 &{}^9C_6 \left( \frac{3}{2} \right)^3 \left( -\frac{1}{3} \right)^6 = {}^9C_3 \times \frac{3^3}{8} \times \frac{1}{3^6} \\
 &= \frac{9 \times 8 \times 7}{3 \times 2} \times \frac{1}{8} \times \frac{1}{3^3} = \frac{7}{18}.
 \end{aligned}$$


---

## Question22

For  $|x| < \frac{1}{\sqrt{2}}$ , the coefficient of  $x$  in the expansion of  $\frac{(1-4x)^2(1-2x^2)^{1/2}}{(4-x)^{3/2}}$  is

**AP EAPCET 2024 - 23th May Morning Shift**

**Options:**

A.  $\frac{61}{64}$

B.  $-\frac{61}{64}$

C.  $\frac{69}{64}$

D.  $-\frac{69}{64}$

**Answer: B**

**Solution:**

To find the coefficient of  $x$  in the expansion of

$$\frac{(1-4x)^2(1-2x^2)^{\frac{1}{2}}}{(4-x)^{\frac{3}{2}}}$$

we proceed by approximating each component in the product.

**Expansion of  $(1-4x)^2$ :**

$$(1-4x)^2 = 1 - 8x + 16x^2$$

**Approximation of  $(1-2x^2)^{\frac{1}{2}}$ :**

Using the binomial approximation for small  $x^2$ , we have:

$$(1-2x^2)^{\frac{1}{2}} \approx 1 - x^2$$

**Approximation of  $(1-\frac{x}{4})^{-\frac{3}{2}}$ :**

Again, using binomial expansion, we write:

$$(1-\frac{x}{4})^{-\frac{3}{2}} \approx 1 + \frac{3}{2} \times \frac{x}{4} = 1 + \frac{3x}{8}$$

Putting it all together, we compute the product:

$$\frac{1}{8} [(1-8x+16x^2)(1-x^2)(1+\frac{3x}{8})]$$

Next, we multiply these expressions and collect the terms involving  $x$ :

$$\begin{aligned} &= \frac{1}{8} \left[ (1-8x+16x^2)(1-x^2)(1+\frac{3x}{8}) \right] \\ &= \frac{1}{8} \left[ (1-8x+16x^2)(1-x^2+\frac{3x}{8}-\frac{3x^3}{8}+\dots) \right] \end{aligned}$$

We are interested only in terms involving  $x$ . Consider:

The coefficient of  $x$  from  $(1 - 8x + 16x^2)(1)$  is  $-8$ .

The contribution due to combining 1 from  $(1 - 8x + 16x^2)$  with the  $\frac{3x}{8}$  term from  $(1 + \frac{3x}{8})$  is  $\frac{3}{8}$ .

Thus, the total coefficient of  $x$  from this product is:

$$\frac{1}{8} \left( \frac{3}{8} - 8 \right)$$

Simplifying, we calculate:

$$\frac{1}{8} \cdot \left( -\frac{61}{8} \right) = -\frac{61}{64}$$

Therefore, the coefficient of  $x$  is  $-\frac{61}{64}$ .

---

## Question23

**If  $P$  is the greatest divisor of  $49^n + 16n - 1$  for all  $n \in N$ , then the number of factors of  $P$  is**

**AP EAPCET 2024 - 22th May Evening Shift**

**Options:**

A. 12

B. 15

C. 7

D. 13

**Answer: C**

**Solution:**

Given that

$P$  is the greatest divisor of  $49^n + 16n - 1$ , for  $\forall$  all  $n \in N$ .

$$\begin{aligned} 49^n + 16n - 1 &= (1 + 48)^n + 16n - 1 \\ &= 1 + n \cdot 48 + \frac{n(n-1)}{2} \cdot (48)^2 + \dots + 16n - 1 \\ &= 64n + {}^nC_2(48)^2 + {}^nC_3(48)^3 + \dots + (48)^n \\ &= 64 \left( n + {}^nC_2 6^2 + {}^nC_3 6^3 \cdot 8 + \dots + 6^n \cdot 8^{n-2} \right) \end{aligned}$$

So, clearly it is divisible by 64 .

Now, number factor of 64 is

$$64 = 2^6 = (6 + 1) = 7$$

---

## Question24

If the coefficients of  $r$  th,  $(r + 1)$  th and  $(r + 2)$  th terms in the expansion of  $(1 + x)^n$  are in the ratio of 4 : 15 : 42, then  $n - r$  is equal to

AP EAPCET 2024 - 22th May Evening Shift

Options:

A. 18

B. 15

C. 14

D. 17

Answer: C

Solution:

Given that coefficient of  $r$  th:  $(r + 1)$  th :  $(r + 2)$  th = 4 : 15 : 42 in expansion of  $(1 + x)^n$ .

$$\begin{aligned}\Rightarrow {}^nC_{r-1} : {}^nC_r : {}^nC_{r+1} &= 4 : 15 : 42 = K \\ \Rightarrow \frac{{}^nC_{r-1}}{{}^nC_r} &= \frac{4K}{15K} \Rightarrow \frac{r}{n-r+1} = \frac{4}{15} \quad \dots (i) \\ \text{and } \frac{{}^nC_r}{{}^nC_{r+1}} &= \frac{15K}{42K} \Rightarrow n-r = \frac{14(r+1)}{5} \quad \dots (ii)\end{aligned}$$

From Eqs. (i) and (ii), we get

$$\begin{aligned}\frac{5r}{14r+19} &= \frac{4}{15} \\ \Rightarrow 75r &= 56r + 76 \Rightarrow 19r = 76 \\ \Rightarrow r &= 4 \\ \Rightarrow n-r &= \frac{14 \times (4+1)}{5} = 14 \text{ [from Eq. (i)]}\end{aligned}$$

---

## Question25

If the coefficients of  $(2r + 6)$  th and  $(r - 1)$  th terms in the expansion of  $(1 + x)^{21}$  are equal, then the value of  $r$  is equal to

AP EAPCET 2024 - 22th May Evening Shift

**Options:**

A. 7

B. 5

C. 6

D. 8

**Answer: C**

**Solution:**

Given that coefficients of  $(2r + 6)$  th and  $(r - 1)$  th terms in the expansion of  $(1 + x)^{21}$  are equal.

$$\begin{aligned}T_{2r+5+1} &= {}^{21}C_{2r+5}(1)^{21-2r-5}x^{2r+5} \\T_{r-2+1} &= {}^{21}C_{r-2}(1)^{21-r+2}x^{r-2} \\ \therefore {}^{21}C_{2r+5} &= {}^{21}C_{r-2} \\ \Rightarrow 2r + 5 &= r - 2 \\ r &= 3 \quad (\text{not possible})\end{aligned}$$

$$\text{So, } 2r + 5 = 21 - (r - 2)$$

$$\begin{aligned}& [\therefore {}^nC_r = {}^nC_{n-r}] \\ \Rightarrow 2r + 5 &= 21 - r + 2 \\ \Rightarrow 3r &= 18 \\ \Rightarrow r &= 6\end{aligned}$$

---

## Question26

If the 2nd, 3rd and 4 th terms in the expansion of  $(x + a)^n$  are 96, 216, 216 respectively and  $n$  is a positive integer, then  $a + x =$

**AP EAPCET 2024 - 22th May Morning Shift**

**Options:**

A.  $n + 1$

B.  $n$

C.  $n - 1$

D.  $\frac{n}{2}$

**Answer: A**

**Solution:**

We know that  $r$  th term in the expansion  $(x + a)^n$  is given by

$$T_{r+1} = {}^nC_r x^{n-r} a^r$$

$$\text{given, } T_2 = {}^nC_1 x^{n-1} a = 96 \quad \dots (i)$$

$$T_3 = {}^nC_2 x^{n-2} a^2 = 216 \quad \dots (ii)$$

$$T_4 = {}^nC_3 x^{n-3} a^3 = 216 \quad \dots (iii)$$

On dividing Eq. (ii) from Eq. (i), we get

$$\frac{{}^nC_2 x^{n-2} a^2}{{}^nC_1 x^{n-1} a} = \frac{216}{96}$$

$$\frac{n!}{(n-2)!2!} x^{n-1-1} a^2$$

$$\frac{n!}{(n-1)!1!} x^{n-1} a = \frac{9}{4}$$

$$\frac{1}{2} \cdot \frac{(n-1)(n-2)!}{(n-2)!} \frac{a}{x} = \frac{9}{4}$$

$$(n-1) \frac{a}{x} = \frac{9}{2}$$

$$(n-1)a = \frac{9}{2}x \quad \dots (iv)$$

Similarly, on dividing Eq. (iii) from Eq. (ii), we get

$$\frac{{}^nC_3 x^{n-3} a^3}{{}^nC_2 x^{n-2} a^2} = \frac{216}{216}$$

$$\Rightarrow \frac{\frac{n!}{(n-3)!3!} x^{n-2-1} a^3}{\frac{n!}{(n-2)!2!} x^{n-2} a^2} = 1$$

$$\Rightarrow \frac{2 \cdot 1(n-2)(n-3)!a}{3 \cdot 2 \cdot l(n-3)!} \frac{a}{x} = 1$$

$$\Rightarrow (n-2)a = 3x \quad \dots (v)$$

Now, on dividing Eq. (v) from Eq. (iv) we get,

$$\frac{(n-2)a}{(n-1)a} = \frac{3x}{\frac{9}{2}x} = \frac{2}{3}$$

$$\Rightarrow 3n - 6 = 2n - 2$$

$$n = 4$$

From Eq. (v), we get

$$(4-2)a = 3x$$

$$a = \frac{3}{2}x \quad \dots (vi)$$

From Eq (i), we get

$${}^4C_1 x^{4-1} a^1 = 96$$

$$\frac{4!}{3!1!} x^3 a = 96$$

$$4x^3 \cdot \frac{3}{2}x = 96 \quad [From Eq. (vi)]$$

$$x^4 = \frac{96}{6} = 16$$

$$x = 2$$

Using Eq. (vi), we get

$$a = 3$$

Clearly,  $x + a = 2 + 3 = 4 + 1$

$$x + a = n + 1$$


---

## Question27

If  $|x| < 1$ , then the number of terms in the expansion of  $\left[\frac{1}{2}(1 \cdot 2 + 2 \cdot 3x + 3 \cdot 4x^2 + \dots \infty)\right]^{-25}$

**AP EAPCET 2024 - 22th May Morning Shift**

**Options:**

A. Infinite

B. 101

C. 76

D. 51

**Answer: C**

**Solution:**

Given the expression:

$$\frac{1}{2}(1 \cdot 2 + 2 \cdot 3x + 3 \cdot 4x^2 + \dots + \infty)^{-25}$$

First, consider the series inside the parentheses:

$$1 \cdot 2 + 2 \cdot 3x + 3 \cdot 4x^2 + \dots + \infty$$

This can be rewritten as:

$$2(1 + 3x + 6x^2 + \dots + \infty)$$

Recognizing the series as a result of differentiating and manipulating a geometric series, it sums to:

$$2 \frac{1}{(1-x)^3}$$

So,

$$\frac{1}{2} \times 2 \times \frac{1}{(1-x)^3} = \frac{1}{(1-x)^3}$$

Thus, the expression evaluates to:

$$\frac{1}{2}(1 \cdot 2 + 2 \cdot 3x + 3 \cdot 4x^2 + \dots + \infty)^{-25} = \left(\frac{1}{(1-x)^3}\right)^{-25} = (1-x)^{75}$$

Therefore, the number of terms in the expansion of  $(1-x)^{75}$  is:

$$75 + 1 = 76$$

---

## Question28

If the ratio of the terms equidistant from the middle term in the expansion of  $(l + x)^{12}$  is  $\frac{1}{256}$  ( $x \in N$ ), then sum of all the terms of the expansion  $(1 + x)^{12}$  is

**AP EAPCET 2024 - 21th May Evening Shift**

**Options:**

A.  $4^{12}$  or  $6^{12}$

B.  $3^{12}$  or  $5^{12}$

C.  $6^{12}$  or  $7^{12}$

D.  $12^{12}$

**Answer: B**

**Solution:**

In the expansion of  $(1 + x)^{12}$ , we need to determine the sum of all its terms given the ratio of terms equidistant from the middle is  $\frac{1}{256}$ .

Let's find the middle term and the equidistant terms:

The  $k$ -th term is denoted by:

$$T_k = \binom{12}{k} x^k$$

The equidistant terms from the middle are:

$$T_{12-k} = \binom{12}{12-k} x^{12-k}$$

Since, the ratio of equidistant terms is given by:

$$\frac{T_k}{T_{12-k}} = \frac{\binom{12}{k} x^k}{\binom{12}{12-k} x^{12-k}} = \frac{1}{256}$$

Since  $\binom{12}{12-k} = \binom{12}{k}$ , the equation simplifies to:

$$\frac{x^k}{x^{12-k}} = x^{-12+2k} = \frac{1}{256} = 2^{-8}$$

This gives us:

$$-12 + 2k = -8$$

Solving for  $k$ , we find:

$$2k = 4 \implies k = 2$$

Thus, the equidistant terms are  $T_2$  and  $T_{10}$ .

Now, calculate the sum of all terms in the expansion:

When  $x = 2$ :

$$S = (1 + 2)^{12} = 3^{12}$$

When  $x = 4$ :

$$S = (1 + 4)^{12} = 5^{12}$$

Therefore, the sum of all terms could either be  $3^{12}$  or  $5^{12}$  based on the value of  $x$ .

---

## Question29

**If the eleventh term in the binomial expansion of  $(x + a)^{15}$  is the geometric mean of the eighth and twelfth terms, then the greatest term in the expansion is**

**AP EAPCET 2024 - 21th May Morning Shift**

**Options:**

A. 7 th term

B. 8 th term

C. 9 th term

D. 10 th term

**Answer: B**

**Solution:**

From the given condition, one  $g^6$

$$\begin{aligned} \left({}^{15}C_{10}x^2a^{10}\right)^2 &= \left({}^{15}C_7x^8a^7\right)\left({}^{15}C_{11}x^4a^{11}\right) \\ \Rightarrow \frac{a}{x} &= \sqrt{\frac{75}{77}} \end{aligned}$$

$$\text{Now, } (x + a)^{15} = x^{15}\left(1 + \frac{a}{x}\right)^{15}$$

Consider the expansion of  $\left(1 + \frac{a}{x}\right)^{15}$

$$T_r = T_{r+1} \Rightarrow \frac{r}{(n - r + 1)\left(\frac{a}{x}\right)} = 1$$

$$\Rightarrow r = 16\sqrt{\frac{75}{77}} - r\sqrt{\frac{75}{77}} \Rightarrow r = \frac{16\sqrt{\frac{75}{77}}}{1 + \sqrt{\frac{75}{77}}}$$

$$r = 7$$

If  $r = 7 \Rightarrow T_7 < T_8$  and  $r = 8 \Rightarrow T_8 > T_9$  Hence,  $T_8$  is the greatest term.

---

## Question30

The sum of the rational terms in the binomial expansion of  $\left(\sqrt{2} + 3^{1/5}\right)^{10}$  is

**AP EAPCET 2024 - 21th May Morning Shift**

**Options:**

A. 41

B. 39

C. 32

D. 30

**Answer: A**

**Solution:**

$(r + 1)$  th term in the given expansion is given by

$$t_{r+1} = {}^{10}C_r (2)^{\frac{10-r}{2}} \cdot 3^{\frac{r}{5}}$$

where  $r = 0, 1, 2, \dots, 10$

For rational terms

$r = a$  multiple of 5 = 0, 5, 10

$10 - r = a$  multiple of 2 = 0, 2, 4

From Eqs. (i) and (ii) possible values of  $r$  are 0 and 10

$\therefore$  Sum of rational terms

$$\begin{aligned} &= t_1 + t_{11} = {}^{10}C_0 (\sqrt{2})^{10} \\ &\quad \left(3^{\frac{1}{5}}\right)^0 + {}^{10}C_{10} (\sqrt{2})^0 \left(3^{\frac{1}{5}}\right)^{10} \\ &= 2^5 + 3^2 = 32 + 9 = 41 \end{aligned}$$

---

## Question31

If the coefficients of  $x^5$  and  $x^6$  are equal in the expansion of  $\left(a + \frac{x}{5}\right)^{65}$ , then the coefficient of  $x^2$  in the expansion of  $\left(a + \frac{x}{5}\right)^4$  is.

**AP EAPCET 2024 - 20th May Evening Shift**

**Options:**

- A. 1
- B.  $\frac{32}{25}$
- C. 2
- D.  $\frac{24}{25}$

**Answer: D**

**Solution:**

Coefficient of  $x^5$  in  $\left(a + \frac{x}{5}\right)^{65}$  is

$${}^{65}C_5 \cdot a^{60} \cdot \left(\frac{1}{5}\right)^5$$

and coefficient of  $x^6$  in  $\left(a + \frac{x}{5}\right)^{65}$  is

$${}^{65}C_6 a^{59} \left(\frac{1}{5}\right)^6$$

Now, coefficient of  $x^5$  = Coefficient of  $x^6$  in expansion  $\left(a + \frac{x}{5}\right)^{65}$

$$\begin{aligned} {}^{65}C_5 \cdot a^{60} \cdot \left(\frac{1}{5}\right)^5 &= {}^{65}C_6 \cdot a^{59} \left(\frac{1}{5}\right)^6 \\ \Rightarrow \frac{65!}{5!60!} \cdot a^{60} \cdot \frac{1}{5^5} &= \frac{65!}{6!59!} a^{59} \cdot \frac{1}{5^6} \\ \Rightarrow \frac{a}{60} &= \frac{1}{6} \times \frac{1}{5} \Rightarrow a = \frac{60}{30} = 2 \\ \Rightarrow a &= 2 \end{aligned}$$

$\therefore$  Coefficient of  $x^2$  in expansion of  $\left(2 + \frac{x}{5}\right)^4$  is

$$\begin{aligned} &= {}^4C_2 \cdot (2)^2 \cdot \left(\frac{1}{5}\right)^2 \\ &= \frac{4!}{2!2!} \cdot \frac{4}{25} = \frac{24}{25} \end{aligned}$$

---

## Question32

If  $|x| < \frac{2}{3}$ , then the 4th term in the expansion of  $(3x - 2)^{\frac{2}{3}}$  is :

# AP EAPCET 2024 - 20th May Evening Shift

**Options:**

A.  $\frac{\sqrt[3]{4}}{6}x^3$

B.  $-\frac{\sqrt[3]{4}}{6}x^3$

C.  $\frac{\sqrt[3]{4}}{8}x^3$

D.  $-\frac{\sqrt[3]{4}}{8}x^3$

**Answer: B**

**Solution:**

Given the expression  $(3x - 2)^{\frac{2}{3}}$ , we can rewrite it as:

$$(3x - 2)^{\frac{2}{3}} = (-2 + 3x)^{\frac{2}{3}}$$

This can be further expressed as:

$$= (2)^{\frac{2}{3}} \left(-1 + \frac{3}{2}x\right)^{\frac{2}{3}}$$

To find the 4th term of the expansion of  $2^{\frac{2}{3}} \left(-1 + \frac{3}{2}x\right)^{\frac{2}{3}}$ , we use the formula for the binomial expansion term, which is given by:

$$T_4 = 2^{\frac{2}{3}} \cdot \binom{\frac{2}{3}}{3} \cdot (-1)^{\frac{2}{3}-3} \cdot \left(\frac{3}{2}x\right)^3$$

Calculating the binomial coefficient:

$$\begin{aligned}\binom{\frac{2}{3}}{3} &= \frac{\frac{2}{3}(\frac{2}{3}-1)(\frac{2}{3}-2)}{3 \times 2 \times 1} \\ &= \frac{\frac{2}{3} \cdot (-\frac{1}{3}) \cdot (-\frac{4}{3})}{6}\end{aligned}$$

Now, simplifying:

$$\begin{aligned}&= \frac{2^{\frac{2}{3}} \cdot (-\frac{2}{27})}{6} \cdot (-1)^{-\frac{7}{3}} \cdot \left(\frac{27}{8}\right)x^3 \\ &= -2^{\frac{2}{3}} \cdot \frac{8}{27} \times \frac{27}{8} \times \frac{1}{6}x^3\end{aligned}$$

Finally, simplifying gives:

$$= -\frac{\sqrt[3]{4}}{6}x^3$$

So, the 4th term in the expansion is  $-\frac{\sqrt[3]{4}}{6}x^3$ .

---

## Question33

The coefficient of  $x^5$  in the expansion of  $\left(2x^3 - \frac{1}{3x^2}\right)^5$  is

**AP EAPCET 2024 - 20th May Morning Shift**

**Options:**

- A. 8
- B. 9
- C.  $\frac{80}{9}$
- D.  $\frac{29}{3}$

**Answer: C**

**Solution:**

We have, to find coefficient of  $x^3$  if the expansion of  $\left(2x^3 - \frac{1}{3x^2}\right)^5$ .

The function is of the type  $\left(ax^p + \frac{b}{x^q}\right)^n$ , then

$$r = \frac{np-k}{p+q} = \frac{5 \times 3 - 5}{3+2} = \frac{5(3-1)}{5} = 2$$

$$\text{Coefficient of } T_{r+1} = {}^nC_r a^{n-r} \cdot b^r$$

$$\text{Coefficient of } T_3 = 5C_2 \cdot (2)^{5-2} \cdot \left(\frac{1}{3}\right)^2$$

$$\begin{aligned} &= \frac{5!}{3! \cdot 2!} \times 2^3 \times \frac{1}{9} \\ &= \frac{5 \times 4}{2} \times 2^3 \times \frac{1}{9} = \frac{80}{9} \end{aligned}$$

-----

## Question34

Numerically greatest term in the expansion of  $(5 + 3x)^6$  When,  $x = 1$ , is

**AP EAPCET 2024 - 19th May Evening Shift**

**Options:**

- A.  $3^5 \times 5^3$
- B.  $3^3 \times 5^5$
- C.  $3^2 \times 5^5$
- D.  $3^4 \times 5^4$

**Answer: B**

## Solution:

To find the numerically greatest term in the expansion of  $(5 + 3x)^6$  when  $x = 1$ , we start by analyzing the general term in the binomial expansion:

$$T_{r+1} = {}^nC_r \cdot x^{n-r} \cdot a^r$$

For our specific case:

$$T_{r+1} = {}^6C_r \cdot 5^{6-r} \cdot (3x)^r$$

We are looking for  $\frac{T_{r+1}}{T_r}$  to determine when the terms start decreasing:

$$\begin{aligned}\frac{T_{r+1}}{T_r} &= \left( \frac{6-r+1}{r} \right) \cdot \frac{3x}{5} \\ &= \frac{21-3r}{5r} \quad (\text{by substituting } x = 1)\end{aligned}$$

Now, we analyze when  $T_{r+1} > T_r$ :

$$\begin{aligned}21 - 3r &> 5r \\ 8r &< 21 \\ r &< 2.625\end{aligned}$$

Thus, to find the numerically largest term, we evaluate  $r$  when  $T_{r+1} > T_r$  and  $T_{r+1} < T_r$ . Specifically, when:

$$\begin{aligned}r < 2.625 : T_{r+1} &> T_r \\ r > \frac{21}{8} : T_{r+1} &< T_r\end{aligned}$$

Now let's calculate  $T_3$  and  $T_4$ :

**For  $T_3$ :**

$$\begin{aligned}T_3 &= {}^6C_2 \cdot 5^4 \cdot (3)^2 \cdot 1^2 \\ &= 15 \cdot 625 \cdot 9 \\ &= 5^5 \cdot 3^3\end{aligned}$$

**For  $T_4$ :**

$$\begin{aligned}T_4 &= {}^6C_3 \cdot 5^3 \cdot (3)^3 \cdot 1^3 \\ &= 20 \cdot 125 \cdot 27 \\ &= 4 \times 5^3 \times 3^3\end{aligned}$$

The calculations show that the largest term is the 3rd term ( $T_3$ ), which equals  $5^5 \times 3^3$ . Thus, the numerically greatest term in the expansion is  $5^5 \times 3^3$ .

-----

## Question35

The square root of independent term in the expansion of  $\left(2x^2 + \frac{5}{x}\right)^5$  is

**AP EAPCET 2024 - 18th May Morning Shift**

**Options:**

A.  $\frac{15}{\sqrt{10}}$

B.  $\frac{10}{\sqrt{15}}$

C.  $\frac{30}{\sqrt{5}}$

D.  $\frac{20}{\sqrt{5}}$

**Answer: C**

**Solution:**

$$T_{r+1} = {}^{36}C_r \left( \frac{2x^2}{5} \right)^{10-r} \times \left( \sqrt{\frac{5}{x}} \right)^r$$

$$\begin{aligned} & {}^{10}C_r \frac{(2)^{32-}}{5^{10-r-\frac{1}{2}}} \times x^{30-3-\frac{r}{2}} \\ &= {}^{10}C_r \frac{(2)^{10-1}}{5^{31-\frac{r}{2}}} \times x^{30-\frac{r}{2}} \end{aligned}$$

Term will be independent if

$$20 - \frac{5r}{2} = 0 \Rightarrow r = 8$$

Independent term

$$\begin{aligned} &= {}^{10}C_3 \frac{(2)^{10-1}}{5} \\ &= {}^{10}C_2 \times 2^2 \times 5^3 = 4500 \end{aligned}$$

$$\text{Square root of } 4500 = 30\sqrt{5}$$

---

## Question36

**The coefficient of  $x^5$  in  $(3 + x + x^2)^6$  is**

**AP EAPCET 2024 - 18th May Morning Shift**

**Options:**

A. 18

B. 540

C. 0

D. 2178

**Answer: D**

**Solution:**

We have,  $(3 + x + x^2)^6$

General term of the expansion

$$= \frac{6!}{r!s!t!} \times 3^r \times x^s \times (x^2)^t$$

Where,

$$r + s + t = 6$$

$$= \frac{6!}{r!s!t!} \times 3^r \times x^{s+2t}$$

For the coefficient of  $x^5$ ,

$$s + 2t = 5$$

$$\text{But } r + s + t = 6$$

$$r + 5 - t = 6$$

$$r - t = 1 \text{ and } s = 5 - 2t$$

Now,  $t = 0$ , then  $r = 1, s = 5$

$t = 1$ , then  $r = 2, s = 3$

$t = 2$  then  $r = 3, s = 1$

$\therefore$  There are only 3 terms that contain  $x^5$

$\therefore$  Coefficient of  $x^5$

$$\begin{aligned} &= \frac{6!}{0!1!5!} \times 3^1 + \frac{6!}{1!2!3!} \times 3^2 + \frac{6!}{2!3!!!} \times 3^3 \\ &= 18 + 540 + 1620 = 2178 \end{aligned}$$

---

## Question37

The absolute value of the difference of the coefficients of  $x^4$  and  $x^6$  in the expansion of  $x^2 - 2x^2 + (x + 1)^4(x^2 - 1)^2$ , is

**AP EAPCET 2024 - 18th May Morning Shift**

**Options:**

A. 13

B. 4

C. 9

D. 1

**Answer: A**

## Solution:

To determine the absolute value of the difference between the coefficients of  $x^4$  and  $x^6$  in the expansion of the given expression, we start with:

$$x^2 - 2x^2 + (x + 1)^4(x^2 - 1)^2$$

We simplify the function as follows:

$$f(x) = \frac{2x^2}{(x^2+1)(x^2+2)}$$

Expanding, we have:

$$f(x) = 2x^2(x^2 + 1)^{-1}(x^2 + 2)^{-1}$$

Simplifying further gives:

$$f(x) = \frac{2x^2}{2} (1 + x^2)^{-1} \left(1 + \frac{x^2}{2}\right)^{-1}$$

This can be expanded into:

$$f(x) = x^2 [1 - x^2 + x^4 - x^6 + \dots] \cdot \left[1 - \frac{x^2}{2} + \frac{x^4}{4} - \frac{x^4}{8} + \dots\right]$$

Now, let's find the coefficients.

For  $x^4$ :

Coefficient calculation:

$$\left[-\frac{1}{2} - 1\right] = \frac{-3}{2}$$

For  $x^6$ :

Coefficient calculation:

$$\left[\frac{1}{4} + \frac{1}{2} + 1\right] = \frac{7}{4}$$

Now calculate the absolute difference:

$$\begin{aligned} \text{Difference} &= \left| \frac{7}{4} - \left(\frac{-3}{2}\right) \right| = \left| \frac{7}{4} + \frac{3}{2} \right| \\ &= \left| \frac{7}{4} + \frac{6}{4} \right| = \left| \frac{13}{4} \right| = \frac{13}{4} \end{aligned}$$

This process allows us to correctly determine the absolute value of the difference between the coefficients of  $x^4$  and  $x^6$  in the expansion.

---

## Question38

The least value of  $n$  so that  ${}^{(n-1)}C_3 + {}^{(n-1)}C_4 > {}^nC_3$

**AP EAPCET 2022 - 4th July Evening Shift**

**Options:**

A. 11

B. 9

C. 8

D. 7

**Answer: C****Solution:**

Given the inequality  $\binom{n-1}{3} + \binom{n-1}{4} > \binom{n}{3}$ :

We start with the given expression:

$$\binom{n-1}{3} + \binom{n-1}{4} > \binom{n}{3}$$

Using the properties of combinations, we know that:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Applying this, we get:

$$\binom{n-1}{3} + \binom{n-1}{4} > \binom{n}{3}$$

This simplifies to:

$$\frac{(n-1)!}{3!(n-4)!} + \frac{(n-1)!}{4!(n-5)!} > \frac{n!}{3!(n-3)!}$$

Rewriting and dividing both sides by common terms:

$$\frac{1}{4!} + \frac{1}{(n-4)} > \frac{1}{n-3 \cdot n-4}$$

Simplifying further:

$$\frac{1}{4} > \frac{1}{(n-3)(n-4)}$$

Taking the reciprocal, we have:

$$4 < (n-3)(n-4)$$

To find the minimum value of  $n$ :

$$(n-3)(n-4) > 4$$

Solving this inequality, we find:

$$n-3 > 4$$

Thus:

$$n > 7$$

The smallest integer  $n$  satisfying the inequality is  $n = 8$ .

Hence, the least value of  $n$  is 8.

---